2D shapes and 3D solids — Area · Surface area · Volume · Units

Remember the units in every answer: cm² for area, cm³ for volume. A calculator is allowed for questions marked □. □ = challenge question.

A · AREA OF 2D SHAPES

1 Work out the area of

a a triangle with base 6 cm and perpendicular height 4.5 cm

b a parallelogram with base 9 cm and perpendicular height 5 cm

c a trapezium with parallel sides 8 cm and 14 cm, and height 5 cm.

2 A shape is made of a rectangle 12 cm wide and 6 cm tall, with a triangle of height 4 cm sitting on its full width. Work out the total area.

3 Calculate the shaded area:

a a square of side 10 cm with a square hole of side 3 cm cut from it

b a rectangle 12 cm by 8 cm with a triangle (base 12 cm, height 8 cm) removed.

4 Two triangles have the same area. The first has base 8 cm and height 6 cm. The second has base 12 cm. Work out its height.

5 A parallelogram has base 6 cm and perpendicular height 35 mm. Work out its area in cm².

B · 3D SOLIDS

6 A cuboid measures 3 cm by 4 cm by 7 cm.

a Work out its volume.

b Work out its surface area.

c It stands on its 7 cm by 4 cm face. Give the dimensions of its plan view, front elevation and side elevation.

7 An L-shaped prism is 8 cm long. Its cross-section is a 10 cm by 2 cm rectangle with a 3 cm by 4 cm rectangle attached. Work out the volume of the prism.

8 ▯ An open box (no lid) is a cuboid 20 cm long, 15 cm wide and 10 cm high. Work out the area of card needed to make it.

9 A cuboid has volume 480 cm³, length 10 cm and height 6 cm. Work out its width.

10 Write down the number of faces, edges and vertices of **a** a triangular prism, **b** a square-based pyramid.

C · UNITS AND PROBLEMS

11 Copy and complete:

a $2.5\ \text{m}^2 = \square\ \text{cm}^2$ **b** $7\,500\ \text{mm}^3 = \square\ \text{cm}^3$ **c** $0.8\ \text{litres} = \square\ \text{cm}^3$

d $3\ \text{cm}^2 = \square\ \text{mm}^2$

12 ▯ Small boxes measure 10 cm by 5 cm by 4 cm. They are packed into a crate 50 cm by 30 cm by 25 cm, with the 10 cm sides along the 50 cm length, the 5 cm sides along the 30 cm width and the 4 cm sides upright.

a How many boxes fit in the crate?

b What volume of the crate is left empty?

13 A trapezium has parallel sides of 9 cm and 15 cm and an area of 72 cm². Work out its height.

Formulas: triangle $\frac{1}{2}bh$ · parallelogram bh · trapezium $\frac{1}{2}(a+b)h$ · cuboid lwh · prism = cross-section × length · surface area = add every face.